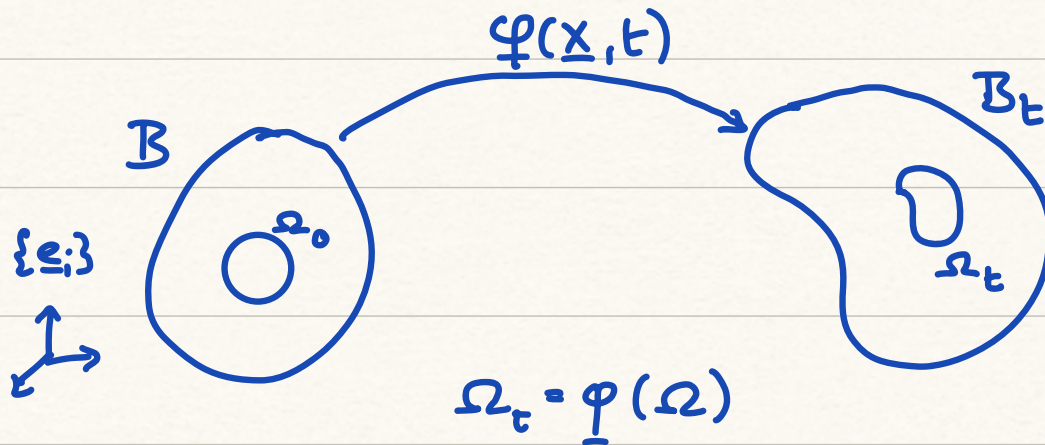


Local Eulerian Balance Laws



Arbitrary subset Ω

I) Conservation of mass

Mass of Ω_t : $M[\Omega_t] = \int_{\Omega_t} \rho(x, t) dV_x$

no reactions or relativistic effects $\Rightarrow$ mass is constant

$$\boxed{\frac{d}{dt} M[\Omega_t] = 0} \quad \Rightarrow \quad M[\Omega_t] = M[\Omega_0]$$

Initial density: $M[\Omega_0] = \int_{\Omega_0} \rho_0(x) dV_x$

Current density: $M[\Omega_t] = \int_{\Omega_t} \rho(x, t) dV_x$

using $dV_x = J(\underline{x}, t) dV_{\underline{x}}$ where $J = \det(\underline{F})$

$$M[\Omega_t] = \int_{\Omega_t} \rho(\underline{x}, t) dV_x = \int_{\Omega_0} \underbrace{\rho(\varphi(\underline{x}, t), t)}_{\rho_m(\underline{x}, t)} J(\underline{x}, t) dV_{\underline{x}}$$

$$= \int_{\Omega_0} \rho_m(\underline{x}, t) J(\underline{x}, t) dV_{\underline{x}}$$

$$\Rightarrow M[\Omega_t] = M[\Omega_0] \Rightarrow \int_{\Omega_0} \rho_m(\underline{x}, t) J(\underline{x}, t) dV_{\underline{x}} = \int_{\Omega_0} \rho_0(\underline{x}) dV_{\underline{x}}$$

$$\int_{\Omega_0} [\rho_m(\underline{x}, t) J(\underline{x}, t) - \rho_0(\underline{x})] dV_{\underline{x}} = 0$$

by arbitrariness of Ω we have

$$\boxed{\rho_m(\underline{x}, t) J(\underline{x}, t) = \rho_0(\underline{x})}$$

Lagrangian statement of mass conservation. ($\underline{x}$)

Convert to Eulerian: $\frac{\partial}{\partial t}$

$$\underbrace{\frac{\partial}{\partial t} \rho_m(\underline{x}, t) J(\underline{x}, t)}_{\dot{\rho}(\underline{x}, t)} + \rho_m(\underline{x}, t) \underbrace{\frac{\partial}{\partial t} J(\underline{x}, t)}_{J(\nabla_{\underline{x}} \cdot \underline{v})_m} = \cancel{\frac{\partial}{\partial t} \rho_0(\underline{x})}^0 = 0$$

dividing by J and switching to spatial description

$$\dot{\rho}(\underline{x}, t) + \rho(\underline{x}, t) \nabla_{\underline{x}} \cdot \underline{v}(\underline{x}, t) = 0$$

$$\Rightarrow \boxed{\dot{\rho} + \rho \nabla_{\underline{x}} \cdot \underline{v} = 0} \quad \text{local Eulerian form}$$

expanding the material derivative we have

$$\frac{\partial \rho}{\partial t} + \underbrace{\nabla_x \rho \cdot \underline{v} + \rho \nabla_x \cdot \underline{v}}_{\nabla_x \cdot (\rho \underline{v})} = 0$$

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla_x \cdot (\rho \underline{v}) = 0} \quad \text{conservative local Eulerian form}$$

Time derivative of integrals relative to mass

$$\boxed{\frac{d}{dt} \int_{\Omega_t} \phi(\underline{x}, t) \rho(\underline{x}, t) dV_x = \int_{\Omega_t} \dot{\phi}(\underline{x}, t) \rho(\underline{x}, t) dV_x}$$

where $\phi(\underline{x}, t)$ is any spatial scalar, vector or tensor field.

$$\int_{\Omega_t} \phi(\underline{x}, t) \rho(\underline{x}, t) dV_x = \int_{\Omega} \phi_m(\underline{x}, t) \underbrace{\rho_m(\underline{x}, t) \det \underline{F}(\underline{x}, t)}_{\rho_0(\underline{x})} dV_x$$

$$\int_{\Omega_t} \phi(\underline{x}, t) \rho(\underline{x}, t) dV_x = \int_{\Omega} \phi_m(\underline{x}, t) \rho_0(\underline{x}) dV_x$$

Take derivative

$$\begin{aligned} \frac{d}{dt} \int_{\Omega_t} \phi(\underline{x}, t) \rho(\underline{x}, t) dV_x &= \int_{\Omega} \frac{d}{dt} \phi_m(\underline{x}, t) \rho_0(\underline{x}) dV_x \\ &= \int_{\Omega} \dot{\phi}_m(\underline{x}, t) \rho_0(\underline{x}) dV_x \\ &= \int_{\Omega} \dot{\phi}_m(\underline{x}, t) \rho_m(\underline{x}, t) \det \underline{F}(\underline{x}, t) dV_x \\ &= \int_{\Omega_t} \dot{\phi}(\underline{x}, t) \rho(\underline{x}, t) dV_x \quad \checkmark \end{aligned}$$

Laws of inertia (Galileo & Newton)

linear momentum: $\underline{L}[\Omega_t] = \int_{\Omega_t} \rho(\underline{x}, t) \underline{v}(\underline{x}, t) dV_x$

angular momentum: $\underline{j}[\Omega_t]_{\underline{z}} = \int_{\Omega_t} (\underline{x} - \underline{z}) \times \rho(\underline{x}, t) \underline{v}(\underline{x}, t) dV_x$

With respect to a fixed frame of reference, the rate of change of linear and angular momentum of any $\Omega_t \subseteq B_t$ equal the resultant force & torque about the origin

$$\frac{d}{dt} \underline{L}[\Omega] = \int_{\Omega_t} \rho(\underline{x}, t) \underline{b}(\underline{x}, t) dV_x + \int_{\partial\Omega_t} \underline{t}(\underline{x}, t) dA_x$$

$$\frac{d}{dt} \underline{j}[\Omega]_o = \int_{\Omega_t} \underline{x} \times \rho(\underline{x}, t) \underline{b}(\underline{x}, t) dV_x + \int_{\partial\Omega_t} \underline{x} \times \underline{t}(\underline{x}, t) dA_x$$

II) Balance of Linear momentum

For an arbitrary $\Omega_t \subseteq B_t$ we have

$$\frac{d}{dt} \int_{\Omega_t} \rho \underline{v} dV_x = \int_{\partial\Omega_t} \underline{t} dA_x + \int_{\Omega_t} \rho \underline{b} dV_x$$

where $\rho, \underline{v}, \underline{t}$ and $\underline{b}$ are spatial fields.

Cauchy stress field: $\underline{t} = \underline{\underline{\sigma}} \underline{n}$

$$\frac{d}{dt} \int_{\Omega_t} \rho \underline{v} dV_x = \int_{\partial\Omega_t} \underline{\underline{\sigma}} \underline{n} dA_x + \int_{\Omega_t} \rho \underline{b} dV_x$$

using tensor divergence theorem

$$\frac{d}{dt} \int_{\Omega_t} \rho \underline{v} dV_x = \int_{\Omega_t} \nabla \cdot \underline{\underline{\sigma}} + \rho \underline{b} dV_x$$

using derivative relative to mass

$$\int_{\Omega_t} \rho \dot{\underline{v}} - \nabla \cdot \underline{\underline{\sigma}} - \rho \underline{b} dV_x = 0$$

by the arbitrariness of Ω_t , we have

$$\boxed{\rho \dot{\underline{v}} - \nabla \cdot \underline{\underline{\sigma}} = \rho \underline{b}} \quad \text{local Eulerian form}$$

Also referred to as Cauchy's first equation of motion.

To rewrite this in conservative form consider the following

$$\rho \dot{\underline{v}} = \rho \frac{\partial \underline{v}}{\partial t} + \rho (\nabla_{\underline{x}} \underline{v}) \underline{v} = \frac{\partial}{\partial t} (\rho \underline{v}) - \frac{\partial \rho}{\partial t} \underline{v} + (\nabla_{\underline{x}} \underline{v}) (\rho \underline{v})$$

using mass balance $\frac{\partial \rho}{\partial t} = -\nabla_{\underline{x}} \cdot (\rho \underline{v})$

$$\rho \dot{\underline{v}} = \frac{\partial}{\partial t} (\rho \underline{v}) + \nabla_{\underline{x}} \cdot (\rho \underline{v}) \underline{v} + (\nabla_{\underline{x}} \underline{v}) (\rho \underline{v})$$

using $\nabla \cdot (\underline{a} \otimes \underline{b}) = (\nabla \underline{a}) \underline{b} + \underline{a} \nabla \cdot \underline{b}$ (see HW5 Q4)

$$\rho \dot{\underline{v}} = \frac{\partial}{\partial t} (\rho \underline{v}) + \nabla_{\underline{x}} \cdot (\rho \underline{v} \otimes \underline{v})$$

Hence we have conservative local Eulerian form

$$\frac{\partial}{\partial t} (\rho \underline{v}) + \nabla_{\underline{x}} \cdot (\rho \underline{v} \otimes \underline{v} - \underline{\underline{\sigma}}) = \rho \underline{b}$$

conserved quantity: $\rho \underline{v}$ = linear momentum

advective mom. flux: $\rho \underline{v} \otimes \underline{v}$

diffusive mom. flux: $-\underline{\underline{\sigma}}$

III, Balance of angular momentum

For an arbitrary $\Omega_t \subseteq B_t$ we have

$$\frac{d}{dt} \int_{\Omega_t} \underline{x} \times \rho \underline{v} dV_x = \int_{\partial\Omega_t} \underline{x} \times \underline{t} dA_x + \int_{\Omega_t} \underline{x} \times \rho \underline{b} dV_x$$

The left hand side becomes

$$\begin{aligned} \frac{d}{dt} \int_{\Omega_t} \rho (\underline{x} \times \underline{v}) dV_x &= \int_{\Omega_t} \rho \frac{d}{dt} (\underline{x} \times \underline{v}) dV_x = \\ &= \int_{\Omega_t} \rho (\dot{\underline{x}} \times \underline{v} + \underline{x} \times \dot{\underline{v}}) dV_x \quad \begin{array}{l} \dot{\underline{x}} = \underline{v} \\ \underline{v} \times \underline{v} = \underline{0} \end{array} \\ &= \int_{\Omega_t} \rho (\underline{x} \times \dot{\underline{v}}) dV_x \end{aligned}$$

Substituting Cauchy stress field the r.h.s becomes

$$\int_{\Omega_t} \rho (\underline{x} \times \dot{\underline{v}}) dV_x = \int_{\partial\Omega} \underline{x} \times \underline{\underline{\sigma}} \underline{n} dA_x + \int_{\Omega_t} \rho (\underline{x} \times \underline{b}) dV_x$$

$$\int_{\Omega_t} \underline{x} \times (\rho \dot{\underline{v}} - \rho \underline{b}) dV_x = \int_{\partial\Omega} \underline{x} \times \underline{\underline{\sigma}} \underline{n} dA_x$$

substitute linear mom. balance $\rho \dot{\underline{v}} - \rho \underline{b} = \nabla_x \cdot \underline{\underline{\sigma}}$

$$\int_{\Omega_t} \underline{x} \times \nabla_x \cdot \underline{\underline{\sigma}} dV_x = \int_{\partial\Omega} \underline{x} \times \underline{\underline{\sigma}} \underline{n} dA_x$$

This is exactly the statement we had for the static case in Lecture 14 on Mechanical Equilibrium.

$\Rightarrow \underline{\underline{\sigma}} = \underline{\underline{\sigma}}^T$ extends to transient cases.